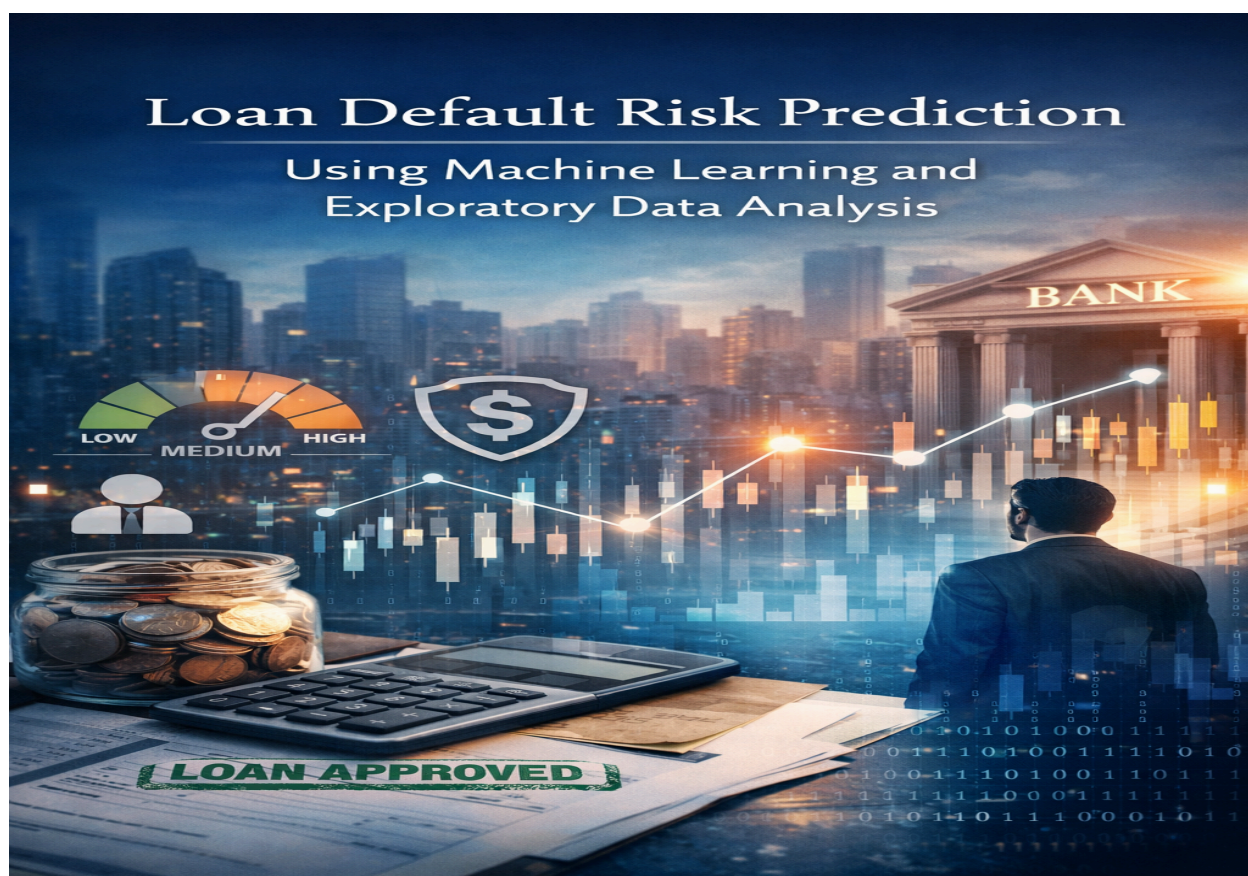




Loan Default Risk Prediction

Using Machine Learning and Exploratory Data Analysis

"A classification model developed to identify high-risk loan applicants and support credit risk decision-making."

Tech Stack

Python • Pandas • NumPy • Matplotlib • Seaborn • Scikit-learn

Key Concepts

EDA • Feature Engineering • Logistic Regression • Model Evaluation

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Project Overview

Financial institutions face significant losses due to loan defaults, making early risk detection essential. The main objective of this project was to perform extensive exploratory data analysis(EDA) to identify patterns associated with loan defaults and to develop a predictive model using logistic regression based on customer financial data and demographic data, and eventually flag high risk customers. The model classifies customers into:

- **0 - Low Risk (Not likely to default)**
- **1 - High Risk (Likely to default)**

The final goal was to develop a machine learning model capable of supporting credit risk assessment by flagging potentially high-risk customers.

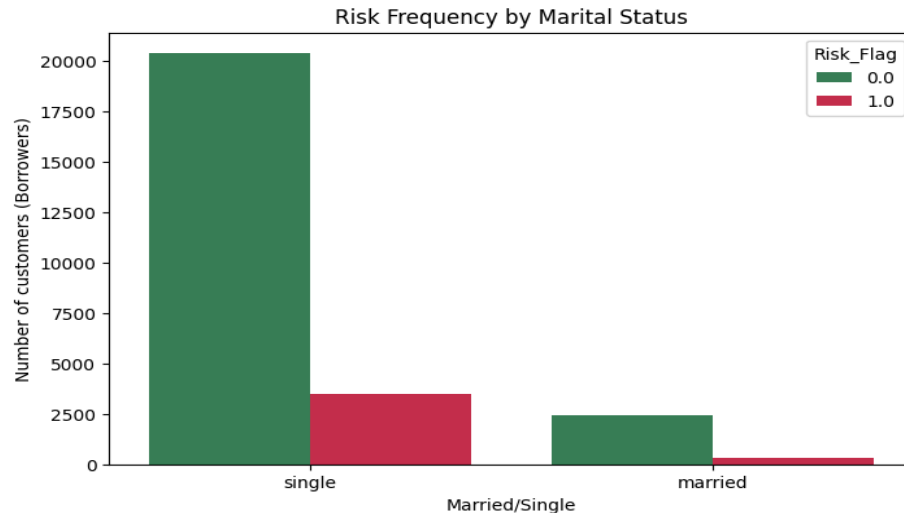
Research Questions and EDA Insights

In order to get a better understanding of the factors influencing loan default risk, several hypothetical questions were formulated and tested using statistical summaries and visualizations. Those hypothetical research questions are

1. Does marital status influence loan default risk?
2. Which age groups are most likely to apply for loans, and does age significantly influence the likelihood of loan default?
3. Does owning an asset like a car affect loan default risk?

EDA Insights

1. Does marital status influence loan default risk?

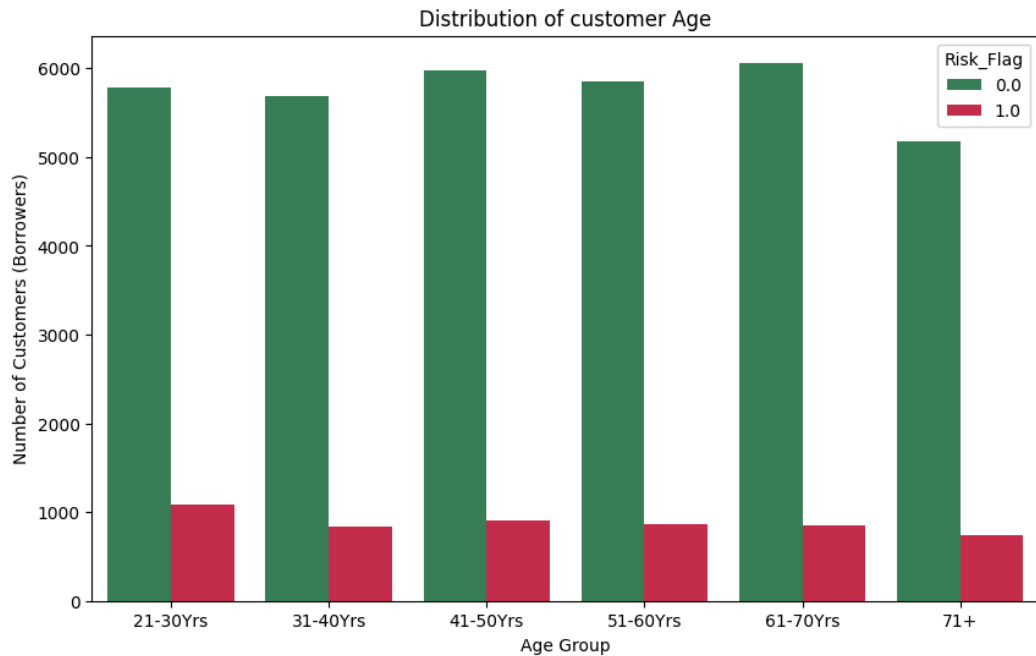


Interpretation: The visualization initially shows that single customers have more loan defaults than married customers, suggesting marital status may influence default risk. However, since there are more single customers overall, the higher number of defaults may simply reflect the larger population. Therefore, the **default rate (proportion)** must be analyzed instead of the total count to accurately determine the true risk difference.

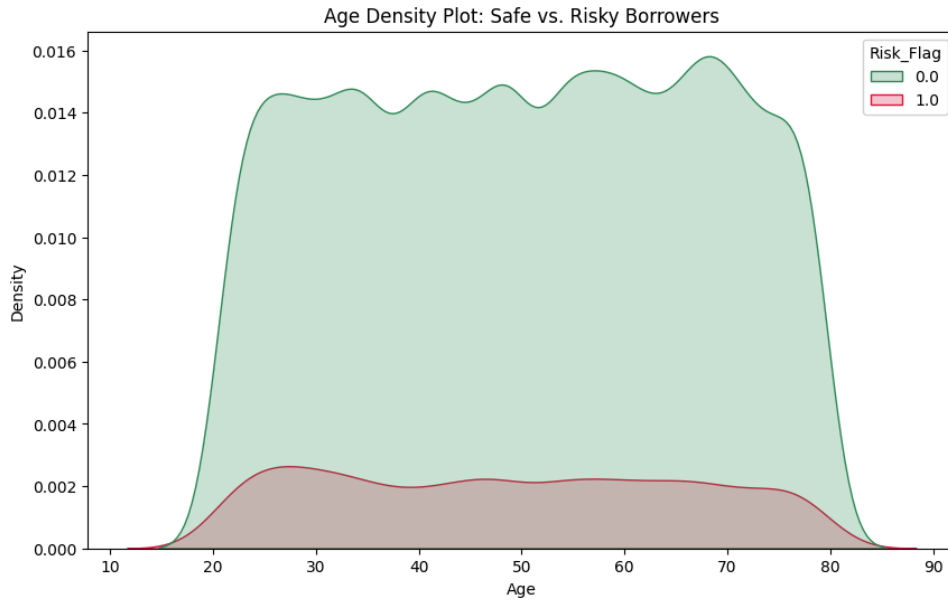
Marital Status	Risk_Flag
married	0.108437 = 10.8%
single	0.135199 = 13.5%

Interpretation: Married people have a lower loan default rate compared to single people who have a loan default rate, therefore marital status has a slight difference on loan defaulting.

2. Which age groups are most likely to apply for loans, and does age significantly influence the likelihood of loan default?

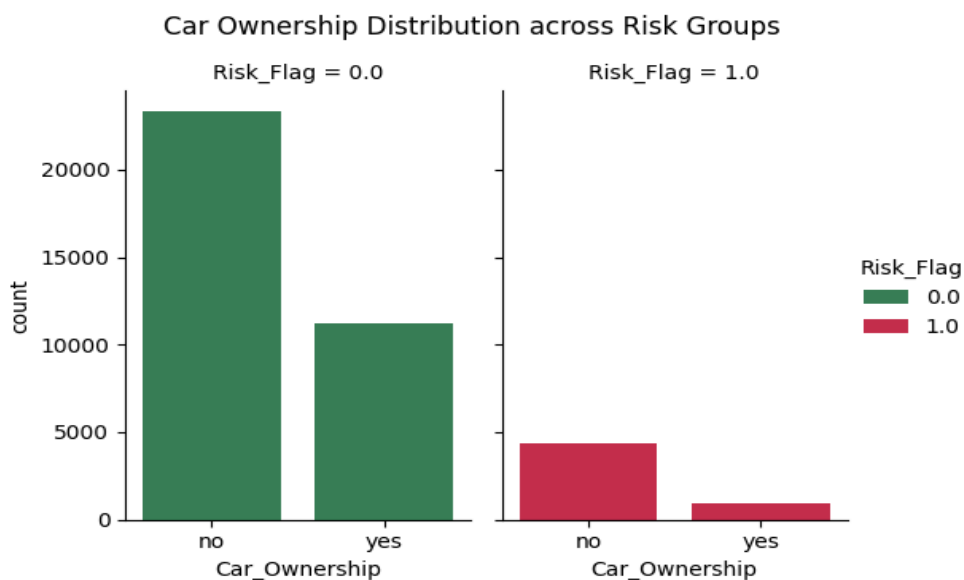


Interpretation: The analysis shows that middle-aged individuals (41–70 years) represent the largest proportion of loan applicants. However, younger customers aged 21–30 exhibit the highest number of loan defaults, suggesting that age may influence loan repayment behavior. This indicates that younger borrowers may present higher credit risk, while older borrowers appear more financially stable. Therefore, age is an important feature to consider in predicting loan default risk.



Interpretation: To add onto the above interpretation, the significant overlap between safe and risky borrower distributions indicates that age alone is not a strong independent predictor of loan default. Therefore, while age contributes useful information, it must be combined with other financial and demographic variables to accurately predict loan risk.

3. Does owning an asset like a car affect loan default risk?



Interpretation: The data is heavily skewed toward low-risk individuals, but the lower ownership rate among high-risk cases suggests a potential correlation between lack of vehicle ownership and higher risk flags.

Car_Ownership	Risk_Flag
no	0.157576 = 15.8%
yes	0.074957 = 7.5%

Interpretation: There is a significant difference in default risk percentage between people who own cars and those who don't by **8.3%**, which implies that car ownership may serve as a proxy for higher disposable income or greater collateral, both of which typically correlate with a lower likelihood of loan defaulting.

Summary: Exploratory data analysis revealed that financial stability indicators are the strongest predictors of loan default risk. Customers who rent homes or lack car ownership exhibit significantly higher default rates, suggesting asset ownership is closely linked to lower risk. Profession-based differences further indicate that employment type influences repayment behavior. While age and marital status show moderate variation, they are less predictive on their own. Overall, income, job stability, and asset ownership are the most influential factors.

Machine Learning Section

- To prepare the data for machine learning, categorical variables were converted into numerical form. Target encoding was applied to **City, Profession, and State**, where each category was replaced with its average risk rate. One-hot encoding was used for **Marital Status and House Ownership** to avoid the model incorrectly treating categories as having a ranking order.

Logistic Regression was chosen as the classification model. Since the dataset was imbalanced, class weighting was applied (`class_weight='balanced'`) to ensure the model paid more attention to identifying high-risk customers.

Model performance was evaluated using accuracy, precision, recall, and the confusion matrix. The results are as follows

Evaluation Metric	Actual Result
Accuracy_Score	63.5%
Precision_Score	21.5%
Recall_Score	61.7%
f1_Score	31.8%

Model Performance Summary

- Accuracy (63.5%)**
The model correctly classified 63.5% of all customers. However, accuracy alone can be misleading due to class imbalance.
- Precision (21.5%)**
Only 21.5% of the customers predicted as high-risk were actually risky. This indicates a relatively high number of false positives.
- Recall (61.7%)**
The model successfully identified 61.7% of actual high-risk customers. This is important because it shows the model is reasonably effective at detecting risky borrowers.
- F1-Score (31.8%)**
The F1-score balances precision and recall. A score of 31.8% indicates that while recall is moderate, low precision reduces overall performance balance.

Confusion Matrix Results

TN = 5832	FP = 3310
FN = 560	TP = 904

- True Negatives (5832): 5,832 safe customers were correctly classified as low-risk.
- False Positives (3310): 3,310 safe customers were incorrectly flagged as high-risk.
- False Negatives (560): 560 risky customers were incorrectly classified as low-risk.
- True Positives (904): 904 risky customers were correctly identified as high-risk.

In summary the model prioritizes catching defaulters, but this comes at the cost of potentially rejecting many creditworthy applicants.

Recommendations

The model can be improved by trying more advanced algorithms (e.g Random Forest) and improving how the data is prepared. Handling class imbalance better may help increase precision. Adding more relevant financial features could also improve predictions. The model should support decision-making, not replace human judgment.

